

Question	Answer	Marks
2(a)	product of mass and velocity	B1
2(b)	change in momentum = $(-1.4) - (+2.8)$ = $(-) 4.2 \text{ kg m s}^{-1}$	A1
2(c)	$F = \Delta p / (\Delta)t$ or $F = \text{gradient}$ = $4.2 / 12$	C1
	= 0.35 N	A1
2(d)	constant / uniform (rate of) decrease (of speed to zero).	B1

Question	Answer	Marks
2(e)	- The (resultant) force is constant / does not decrease - air resistance would vary / decrease / not constant - the force is not zero when speed / velocity is zero / at 8 s Any two of the above 3 marking points (1 mark each, max 2)	B2
2(f)	line from origin with decreasing positive gradient	B1
	gradient changes from positive to negative at 8.0 s	B1
	after $t = 8.0$ s the line has a negative gradient of increasing magnitude <u>and</u> a positive value of d at $t = 12$ s	B1

Question	Answer	Marks
3(a)	1	C1